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# Theme 3.1 — Business Growth and Business Objectives

## 1. Sizes of Firms

The **size of a firm** can be measured by: number of employees, turnover (revenue), market capitalisation, profit, or market share. Different measures can give different rankings.

| Classification | Employees | Turnover | Examples |
|---|---|---|---|
| Micro | 0–9 | < £632k | Local café, freelance web designer |
| Small | 10–49 | < £10.2m | Independent restaurant chain, craft brewery |
| Medium | 50–249 | < £36m | Regional law firm, mid-size manufacturer |
| Large | 250+ | ≥ £36m | Tesco, Amazon, Unilever |

**Why do small firms survive?**

- They serve **niche markets** too small to attract large firms
- They offer **personalised services** (e.g. bespoke tailoring, local plumbing)
- They operate as **suppliers to large firms** in supply chains
- **Barriers to growth** — lack of finance, limited management capacity
- The owner may **prefer to stay small** (lifestyle business, satisficing)
- Some industries have **low minimum efficient scale** (hairdressing, consultancy)

**Exam tip:** When asked about firm size, always specify *which measure* you are using. A firm can be "large" by revenue but "small" by employees (e.g. a highly automated tech company).

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# 2. Why Firms Grow

Firms seek to grow for several strategic and economic reasons:

- **Economies of scale** — lower long-run average costs as output increases
- **Increased market power** — ability to set higher prices, influence suppliers
- **Increased market share** — greater dominance and pricing power
- **Diversification** — spread risk across products/markets
- **Managerial motives** — empire building, higher salaries linked to firm size
- **Shareholder pressure** — growth increases share price and dividends
- **Survival** — in competitive markets, firms that don't grow may be acquired or fail
- **Synergies** — combining complementary resources from two firms

**Constraints on growth:**

- **Size of the market** — limited demand caps potential growth
- **Access to finance** — banks may refuse lending; small firms face credit constraints
- **Diseconomies of scale** — costs rise if the firm grows too large
- **Regulation** — Competition and Markets Authority (CMA) may block mergers
- **Owner objectives** — some owners prefer satisficing over maximising growth
- **Management capacity** — shortage of skilled managers limits coordination

# 3. Organic vs External Growth

## 3a. Internal (Organic) Growth

> **Organic growth** is expansion from within the firm, using its own resources: reinvesting profits, expanding capacity, developing new products, hiring more workers, or entering new markets.

**Advantages:** Lower risk, maintains corporate culture, financed from retained profits, steady and sustainable.

**Disadvantages:** Slow, may miss market opportunities, limited by internal resources.

## 3b. External Growth (Mergers and Acquisitions)

**External growth** occurs through mergers (two firms agree to join) or acquisitions/takeovers (one firm buys another). It allows rapid expansion and access to new markets, technologies, or resources.

| Type of Integration | Description | Motives | Example |
|---|---|---|---|
| **Horizontal** | Merger between firms at the same stage of production in the same industry | Economies of scale, increased market share, eliminate competitor | Facebook acquiring Instagram (2012) |
| **Vertical forward** | Merger with a firm at a later stage (closer to consumer) | Control distribution, secure retail outlets, higher profit margins | Brewery buying a pub chain |
| **Vertical backward** | Merger with a firm at an earlier stage (closer to raw materials) | Secure supply, control quality, reduce input costs | Supermarket buying a farm |
| **Conglomerate (lateral)** | Merger with a firm in a completely different industry | Diversification, risk-bearing economies, asset stripping | Virgin Group (airlines, trains, media, health) |

**Evaluation — Are mergers always beneficial?**
**Benefits:** Economies of scale, synergies, access to new markets, greater R&D capacity.
**Costs:** Diseconomies of scale, culture clashes, job losses, reduced competition and choice for consumers, potential for regulatory intervention by the CMA.
**Key point:** Many mergers fail to deliver expected synergies — studies suggest over 50% of mergers destroy shareholder value.

### Case Study: Facebook/Meta's Acquisition of Instagram (2012) and WhatsApp (2014)

Facebook acquired Instagram for $1bn and WhatsApp for $19bn. These were **horizontal integrations** in the social media market. Motives included: eliminating potential competitors, acquiring user bases (Instagram had 30m users; WhatsApp had 450m), and gaining access to

mobile-first platforms. Critics argued these acquisitions reduced competition and contributed to Meta's dominance, leading to FTC antitrust action in the US. This illustrates how external growth can increase **monopoly power** and reduce **contestability**.

# 4. Economies and Diseconomies of Scale

## 4a. Economies of Scale

**Economies of scale** are reductions in long-run average cost (LRAC) as a firm increases its scale of production. They cause the downward-sloping section of the LRAC curve.

**Internal economies of scale** (arise from the growth of the firm itself):

| Type | Explanation | Real-World Example |
|---|---|---|
| **Technical** | Larger firms use more efficient, specialised capital equipment. The *container principle*: doubling the surface area of a container more than doubles its volume. *Indivisibilities*: expensive machinery can only be justified at high output levels. | Amazon's automated fulfilment centres with Kiva robots reduce per-unit handling costs |
| **Purchasing (bulk-buying)** | Large orders give bargaining power to negotiate discounts from suppliers, reducing unit input costs. | Tesco negotiates lower wholesale prices than a corner shop |
| **Managerial** | Large firms can employ specialist managers (finance director, HR director), increasing efficiency. Division of labour at management level. | Amazon employs specialist logistics, AI, and cloud computing teams |
| **Financial** | Large firms access cheaper finance — lower interest rates, ability to issue bonds/shares, more collateral for loans. Perceived as lower risk by lenders. | Apple can borrow at near-government rates due to its $200bn+ cash reserves |

| | | |
|---|---|---|
| **Marketing** | Marketing and advertising costs are spread over a much larger output. A TV advert costs the same whether a firm sells 1,000 or 1,000,000 units. | Coca-Cola's global advertising budget is huge in total but tiny per unit sold |
| **Risk-bearing** | Large firms can diversify across products, markets, and geographies, reducing the impact of a downturn in any one area. | Unilever sells 400+ brands across food, cleaning, and personal care |

**External economies of scale** (arise from the growth of the industry or geographic cluster):

- **Skilled labour pool** — clustering of firms creates a pool of specialised workers (e.g. tech workers in Silicon Valley, financial services in the City of London)
- **Specialist suppliers** — component suppliers and support services locate near large industries
- **Knowledge and technology spillovers** — ideas and innovations spread between nearby firms (e.g. university research parks)
- **Infrastructure improvements** — government invests in transport/broadband to support growing industries

## 4b. Diseconomies of Scale

**Diseconomies of scale** occur when a firm becomes so large that LRAC begins to rise. They cause the upward-sloping section of the LRAC curve.

**Causes:**

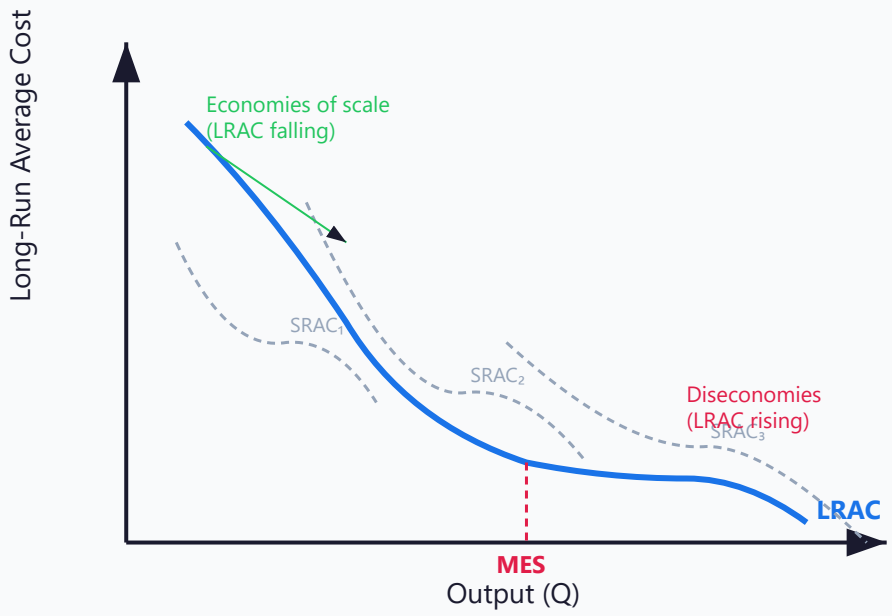
- **Communication problems** — messages distorted through many management layers; slow decision-making
- **Coordination difficulties** — harder to manage and organise a large, complex organisation
- **Loss of control** — senior managers cannot monitor all workers; shirking increases
- **Worker alienation** — employees feel like a small cog; motivation and productivity fall
- **Bureaucracy** — excessive paperwork, rules, and procedures slow down operations
- **Principal-agent problem** — managers may pursue their own goals rather than cost minimisation

## 4c. Minimum Efficient Scale (MES)

**Minimum Efficient Scale (MES)** is the lowest level of output at which long-run average costs are minimised. It is the point where all significant internal economies of scale have been exhausted.

The position of MES relative to market demand determines market structure:

- **High MES relative to market size** → few firms can operate efficiently → oligopoly or natural monopoly (e.g. water supply, railways)
- **Low MES relative to market size** → many firms can coexist → competitive markets (e.g. restaurants, hairdressing)




*The LRAC curve is the envelope of SRAC curves. LRAC falls due to economies of scale, reaches its minimum at MES, and may rise due to diseconomies of scale.*

### Case Study: Amazon's Economies of Scale

Amazon is a textbook example of economies of scale driving competitive advantage. **Technical economies:** Amazon operates 175+ fulfilment centres worldwide, using robotics and AI to reduce per-unit handling costs. **Purchasing economies:** Its scale allows it to negotiate the lowest wholesale prices from publishers and suppliers. **Financial economies:** Amazon raised billions through bond issues at low interest rates to fund expansion (e.g. AWS infrastructure). **Managerial economies:**

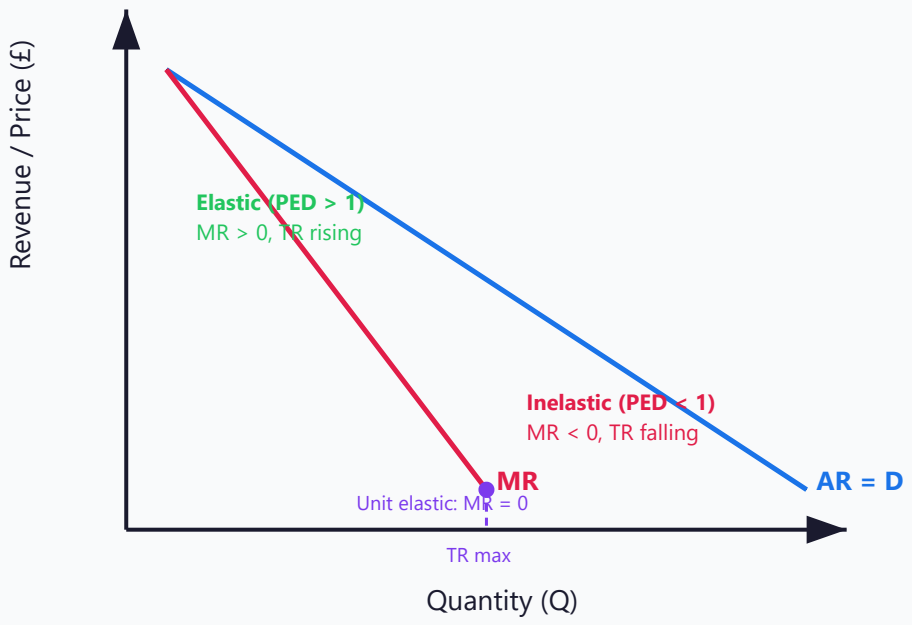
Specialist teams manage logistics, cloud computing, and advertising separately. These economies of scale create a significant **barrier to entry** — a new entrant cannot match Amazon's unit costs without similar scale.

## 5. Revenue

**Total Revenue (TR)** = Price × Quantity = P × Q
**Average Revenue (AR)** = TR ÷ Q = Price. The AR curve *is* the demand curve.
**Marginal Revenue (MR)** = change in TR from selling one additional unit = ΔTR / ΔQ

**For a price-taking firm** (perfect competition): the firm faces a perfectly elastic (horizontal) demand curve, so AR = MR = Price at all output levels.

**For a price-making firm** (monopoly, oligopoly): the demand curve slopes downward. To sell more, the firm must lower its price on *all* units, so MR falls faster than AR. Mathematically, if the AR (demand) curve is linear, the MR curve has the same vertical intercept but twice the gradient — it bisects the horizontal distance between the price axis and the demand curve.




*For a price-maker, MR has twice the gradient of AR. TR is maximised where MR = 0 (unit elastic point). In the elastic region MR > 0; in the inelastic region MR < 0.*

**Relationship between TR and PED:**

| PED Region | MR | Effect of Price Cut on TR |
|---|---|---|
| Elastic (\|PED\| > 1) | MR > 0 | TR increases (% rise in Q > % fall in P) |
| Unit elastic (\|PED\| = 1) | MR = 0 | TR is maximised |
| Inelastic (\|PED\| < 1) | MR < 0 | TR falls (% rise in Q < % fall in P) |

**Exam tip:** A rational firm will never produce in the inelastic section of its demand curve — it could raise price, sell fewer units, and increase total revenue while also reducing total costs.

# 6. Costs of Production

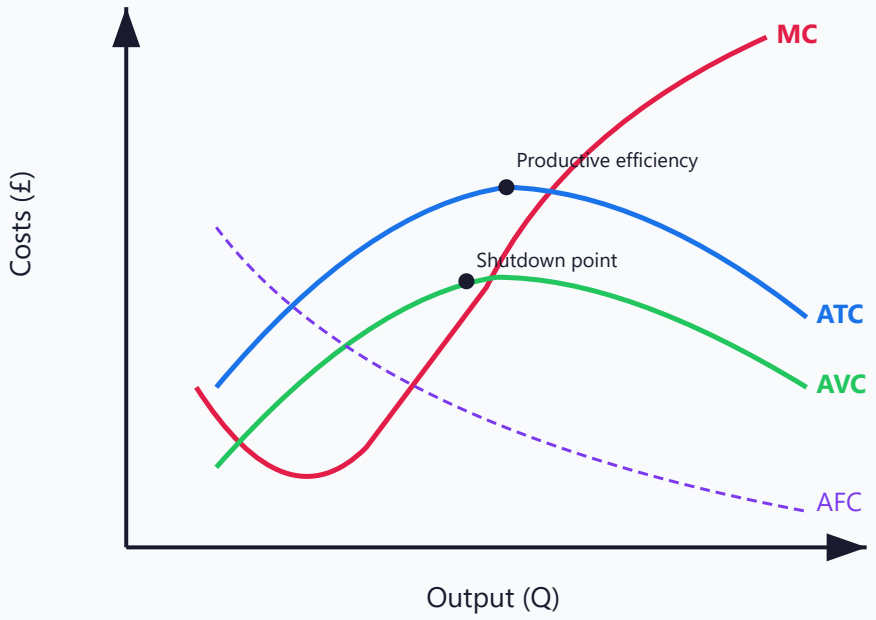
## 6a. Short-Run Costs

In the **short run**, at least one factor of production is fixed (e.g. capital, factory size). The firm can only vary some inputs (e.g. labour, raw materials).

**Fixed Costs (FC)** — do not vary with output in the short run (rent, insurance, salaries of permanent staff, loan repayments)
**Variable Costs (VC)** — vary directly with output (raw materials, energy, piece-rate wages, packaging)
**Total Cost (TC)** = FC + VC
**Average Fixed Cost (AFC)** = FC ÷ Q — always falls as output increases (spreading the overhead)
**Average Variable Cost (AVC)** = VC ÷ Q
**Average Total Cost (ATC or AC)** = TC ÷ Q = AFC + AVC
**Marginal Cost (MC)** = ΔTC / ΔQ = the cost of producing one additional unit

**Law of Diminishing Marginal Returns (LDMR):** In the short run, as successive units of a variable factor (e.g. labour) are added to at least one fixed factor (e.g. capital), there comes a point beyond which the marginal product (MP) of the variable factor begins to decline.

This is the fundamental reason why MC eventually rises and gives short-run cost curves their characteristic U-shape. When MP is rising, MC is falling; when MP is falling, MC is rising.




*Short-run cost curves. MC crosses AVC and ATC at their respective minimum points. The gap between ATC and AVC equals AFC, which falls continuously. Productive efficiency is at minimum ATC.*

**Key relationships:**

- **MC crosses AVC at the minimum of AVC** — this is the *shutdown point* in the short run (if P < AVC, the firm should cease production)
- **MC crosses ATC at the minimum of ATC** — this is the point of *productive efficiency*
- When MC < ATC, ATC is falling; when MC > ATC, ATC is rising (the "marginals pull the average" rule)
- The U-shape of MC, AVC, and ATC is caused by **diminishing marginal returns**
- AFC always declines — fixed costs are spread over more units

**Exam tip — Short-run shutdown rule:** A firm should continue producing in the short run as long as P ≥ AVC (i.e. revenue covers variable costs and makes *some* contribution to fixed costs). If P < AVC, the firm should shut down immediately as it loses more by producing than by stopping.

## 6b. Short Run vs Long Run

**Short run:** The period in which at least one factor of production is fixed. The firm can vary output by changing variable factors only.
**Long run:** The period in which *all* factors of production are variable. The firm can change its scale of production (e.g. build a new factory, buy new machinery).
**Note:** The short run and long run are not defined by calendar time — they depend on the industry. For a software firm, the "long run" may be months; for a nuclear power station, it could be decades.

### 6c. Long-Run Average Cost (LRAC) Curve

In the long run, the firm chooses the optimal combination of *all* factors for each level of output. The **LRAC curve** is the *envelope* of all possible SRAC curves — it traces the lowest possible average cost at each output level.

- **Downward-sloping section:** Economies of scale — LRAC falls as output increases
- **Flat section:** Constant returns to scale — LRAC is constant
- **Upward-sloping section:** Diseconomies of scale — LRAC rises
- The minimum point of the LRAC is the **Minimum Efficient Scale (MES)**

**Exam tip:** Do not confuse *diminishing returns* (a short-run concept caused by a fixed factor) with *diseconomies of scale* (a long-run concept where all factors are variable but coordination problems arise). They are conceptually different.

## 7. Profit

**Profit** = Total Revenue − Total Cost (including opportunity costs)

**Normal profit** — the minimum return required to keep a firm in the industry in the long run. It is the *opportunity cost of enterprise* — the return the entrepreneur could earn in their next best alternative. Occurs when TR = TC (economic profit = 0). Normal profit is *included* in total costs.

**Supernormal (abnormal) profit** — any profit above normal profit. Occurs when TR > TC. Acts as a *signal* for new firms to enter the market.

**Subnormal profit (economic loss)** — profit below the normal level. TR < TC. In the long run, the firm should exit the industry. In the short run, the firm may continue if P ≥ AVC.

$$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$$

$$\text{Supernormal Profit} = (\text{AR} - \text{ATC}) \times \text{Q}$$

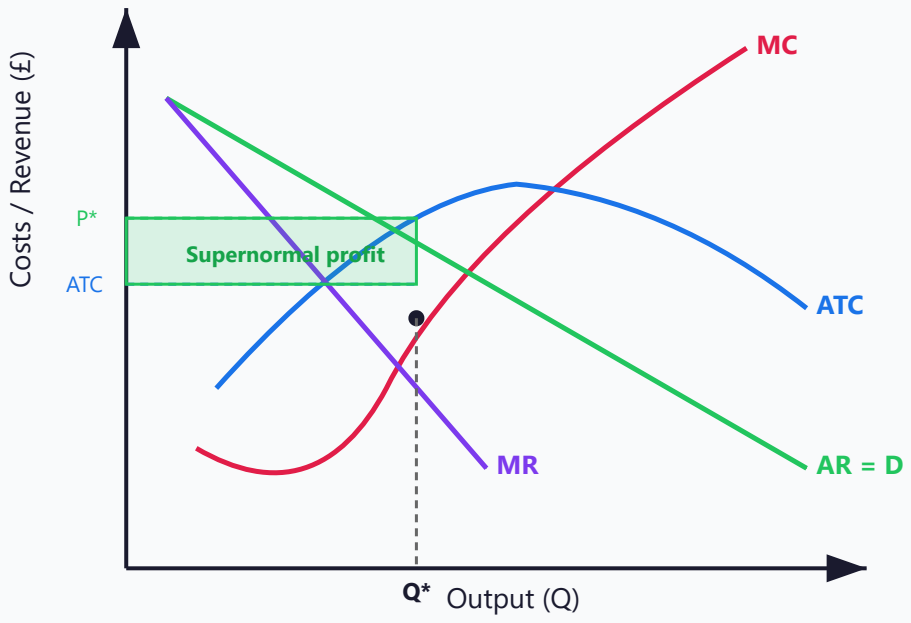
**Exam tip:** Remember that economists include *opportunity costs* in total cost. This means "normal profit" is built into TC. When an economist says "zero profit," they mean the firm is covering all its costs including the opportunity cost of the entrepreneur's time and capital — it is still making an accounting profit.

## 8. Profit Maximisation (MC = MR)

**Profit maximisation** occurs at the output level where **MC = MR** (and MC is rising through MR, or MC cuts MR from below). At this point, the cost of producing one more unit exactly equals the revenue gained from selling it — there is no further profit to be made from additional output.

### Why MC = MR?

- If MR > MC: producing one more unit adds more to revenue than to cost → profit increases → the firm should expand output
- If MR < MC: producing one more unit adds more to cost than to revenue → profit decreases → the firm should reduce output
- Only at MR = MC is there no incentive to change output → profit is maximised




*Profit maximisation at MC = MR. The firm produces Q* and charges P* (read from the AR curve). Supernormal profit = (P* − ATC) × Q*, shown by the shaded rectangle.*

**Criticisms of the profit maximisation assumption:**

- Firms may lack the **perfect information** needed to identify the MC = MR point precisely
- The **principal-agent problem** means managers may not pursue profit maximisation
- Firms may prioritise **long-run growth** over short-run profit (e.g. Amazon ran losses for years to build market share)
- In practice, firms often use **cost-plus pricing** (mark-up on average cost) rather than marginal analysis

# 9. Alternative Business Objectives

## 9a. Revenue Maximisation

**Revenue maximisation** occurs where **MR = 0** (total revenue is at its peak). Output is higher and price is lower than under profit maximisation.

**Rationale:** Managers may be rewarded based on sales revenue rather than profit. It can also be used as a predatory strategy to gain market share, or to build brand loyalty through lower prices.

### 9b. Sales Maximisation (Baumol, 1959)

> **Sales maximisation** means the firm maximises output (units sold) subject to a **profit constraint** — the firm must still earn at least normal profit (AC = AR). Output is higher and price lower than both profit maximisation and revenue maximisation.

**Rationale:** Managerial prestige is linked to firm size. Larger firms are harder to take over. Greater market share deters entry by rivals.

### 9c. Satisficing (Herbert Simon)

> **Satisficing** means the firm aims for a *satisfactory* level of profit (or other objectives) rather than the maximum. The firm "satisfices" when it achieves outcomes acceptable to all key stakeholders (shareholders, managers, workers, customers).

This is closely linked to the **principal-agent problem**: managers (agents) may not maximise profits for shareholders (principals) because they have different objectives. Satisficing represents a compromise between competing stakeholder demands.

### 9d. Survival

In the short run, especially during recessions or for new firms, the primary objective may simply be **survival**. The firm may accept losses and cut prices just to remain in business, hoping conditions improve. This is particularly relevant during economic downturns, pandemics, or when facing aggressive competition.

### 9e. Social Enterprise / Ethical Objectives

> **Social enterprises** are businesses that trade to address social or environmental problems. They reinvest profits back into their social mission rather than maximising returns to shareholders. Examples: The Big Issue, TOMS Shoes, fair-trade cooperatives.

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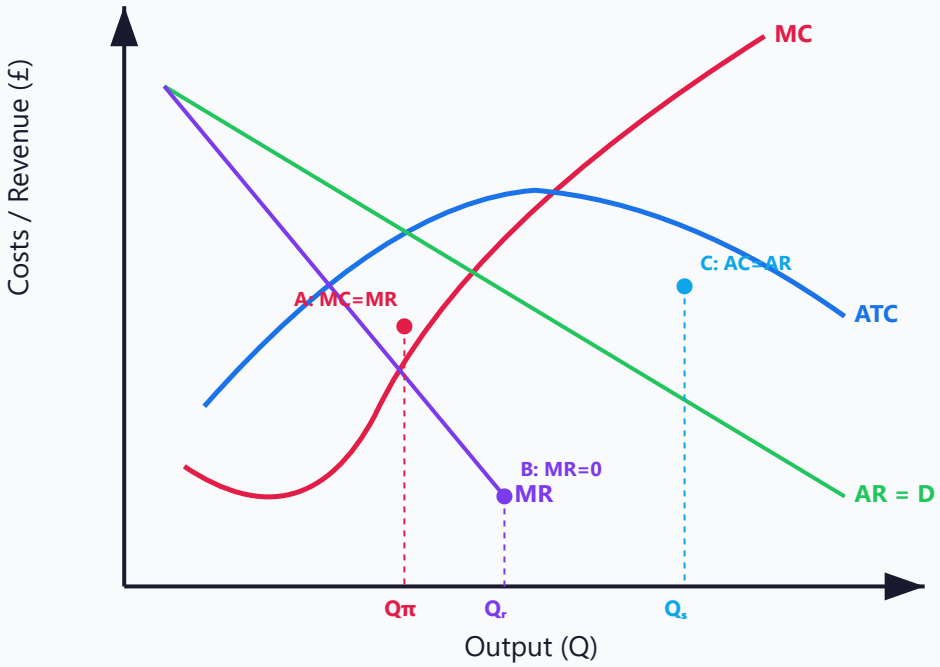
Some for-profit firms also adopt **corporate social responsibility (CSR)** objectives — reducing environmental impact, treating workers fairly, supporting communities — though critics argue this is often "greenwashing" that still serves profit maximisation in the long run through brand enhancement.

## 9f. Comparison of All Objectives




*Comparing objectives: A = Profit maximisation (MC = MR, lowest output, highest price). B = Revenue maximisation (MR = 0). C = Sales maximisation (AC = AR, highest output, lowest price).*

| Objective | Condition | Output | Price | Who Prefers It? |
|---|---|---|---|---|
| Profit maximisation | MC = MR | Lowest | Highest | Shareholders (neoclassical theory) |
| Revenue maximisation | MR = 0 | Medium | Medium | Managers paid on revenue |
| Sales maximisation | AC = AR (normal profit) | Highest | Lowest | Managers seeking empire |

| | | | | |
|---|---|---|---|---|
| Satisficing | Acceptable profit level | Variable | Variable | Managers balancing stakeholders |
| Survival | P ≥ AVC (short run) | Variable | Variable | Firms in distress |

## 10. Principal-Agent Problem and Divorce of Ownership and Control

**Principal-agent problem:** A conflict of interest that arises when one party (the *agent*, e.g. a manager) makes decisions on behalf of another party (the *principal*, e.g. a shareholder). The agent may pursue their own interests (higher salary, perks, empire building, leisure) rather than the principal's interest (profit maximisation).

**Divorce of ownership and control:** In large public limited companies (PLCs), ownership is dispersed among thousands of shareholders, while day-to-day control is exercised by professional managers. Shareholders lack the information and expertise to monitor managers effectively, creating scope for agents to pursue their own objectives.

**Why does this problem arise?**

- **Asymmetric information** — managers know more about the firm's operations than shareholders
- **Dispersed ownership** — individual shareholders have little incentive to monitor (free-rider problem)
- **Complexity** — shareholders cannot easily assess whether managers are maximising profit or shirking

**Possible solutions:**

- **Performance-related pay** — bonuses linked to profit targets, share prices
- **Share options** — giving managers shares so their wealth is tied to shareholder value
- **Threat of takeover** — underperforming firms are bought by rivals who replace managers
- **Non-executive directors** — independent board members who monitor management
- **Regulation and transparency** — mandatory financial reporting, corporate governance codes

**Evaluation — Does the principal-agent problem mean firms never profit maximise?**
Not necessarily. Competitive product markets, the threat of takeover, and well-designed incentive

contracts can push managers towards profit maximisation. However, in firms with significant monopoly power and dispersed ownership, the problem is most acute. The degree of the problem varies by **market structure** — in highly competitive markets, firms that don't profit-maximise are driven out, disciplining managers.

### Case Study: John Lewis Partnership

The John Lewis Partnership is owned by its employees ("Partners"), who share in profits through an annual bonus and have a say in management decisions through elected councils. This **mutual ownership model** reduces the principal-agent problem because workers are both agents *and* principals. Partners have a direct incentive to work hard and keep costs low because they benefit from higher profits. However, critics note that the model limits access to external capital (no shares to sell to investors), potentially constraining growth. The Partnership's bonus has varied significantly — from 18% of salary in 2011 to 2% in 2023 — reflecting the challenge of competing with PLCs that can raise equity finance more easily. This case illustrates how **ownership structure** affects business objectives and the severity of the principal-agent problem.

## Key Formulae Summary

| Concept | Formula |
|---|---|
| Total Revenue | $TR = P \times Q$ |
| Average Revenue | $AR = TR \div Q = P$ |
| Marginal Revenue | $MR = \Delta TR / \Delta Q$ |
| Total Cost | $TC = FC + VC$ |
| Average Total Cost | $ATC = TC \div Q = AFC + AVC$ |
| Average Fixed Cost | $AFC = FC \div Q$ |
| Average Variable Cost | $AVC = VC \div Q$ |
| Marginal Cost | $MC = \Delta TC / \Delta Q$ |

| | |
|---|---|
| Profit | $\pi = TR - TC$ |
| Supernormal Profit | $(AR - ATC) \times Q$ |
| Profit Maximisation | MC = MR (MC rising) |
| Revenue Maximisation | MR = 0 |
| Sales Maximisation | AC = AR (normal profit constraint) |

**Exam tip — Essay structure for 25-mark questions on business objectives:**

1. Define the objective and show it on a diagram (MC = MR for profit max).
2. Explain *why* a firm might pursue this objective (link to market structure, ownership).
3. Evaluate using the principal-agent problem — do firms actually have the information and incentive to maximise profit?
4. Use real-world examples to support analysis (Amazon, John Lewis, etc.).
5. Conclude: the objective pursued depends on market structure, ownership, competition, and time horizon.

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